0.1 Jacobi Method

- **Linear system:** $Ax = b$
- **Residual:** $r = b - Ax_k$
- **Error:** $e = x - x_k$
- **Preconditioner:** $P \approx A$

The Jacobi method is a simple iterative method for solving linear systems of equations. It is a first-order method, and is given by the following update rule:

$$x^{k+1} = D^{-1}(b - (L + U)x^k)$$

where D is the diagonal of the matrix A, L is the lower triangular part of A, and U is the upper triangular part of A.

Source:

- *A Multigrid Tutorial* by Briggs, et al. (2000)
- https://yaningliucudenver.github.io/Numerical-Analysis-I/bookchapter3-2.html